BINGO VINTAGE

HandzJ DNA Refinement & Production Verification Report

Not a redesign. Evidence-based verification of existing implementation.

Composition date: 30 October 2026

Classification key: **VERIFIED** (inspected and/or tested here) · **IMPLEMENTED BUT NOT VERIFIED** (in source; runtime not proven in this environment) · **NOT IMPLEMENTED** · **BLOCKED**.

1. Phase 1 — Baseline preserved

Latest tree under bingo_vintage_complete retains role-specific dashboard, GuidedTour, reports module with Excel exports, migration 0029, seed scripts, and CHANGE_LOG_HANDZJ_DASHBOARD.md. No rebuild of Nest/Next architecture.

Item	Classification
Role-specific dashboard in page.tsx	IMPLEMENTED BUT NOT VERIFIED (runtime on your host)
Reports six tabs + Excel	IMPLEMENTED BUT NOT VERIFIED
Guided tour	IMPLEMENTED BUT NOT VERIFIED
Double /api path fix	IMPLEMENTED BUT NOT VERIFIED

2. Phase 2 — Interface DNA pass

Dashboard and Reports received HandzJ anti-template treatment. Selective DNA across Clients, Loans, Payments, Drawers, Expenses form/table systems is **NOT IMPLEMENTED** as a full pass in this phase (would be multi-file operational UI work). Recommendation: one module per iteration without changing business rules.

3. Phase 3 — NPL and ageing verification (source)

loans.balance: used as remaining principal outstanding in portfolio summary and NPL ratio (not full contractual interest). **total_arrears** / **days_in_arrears**: written by ArrearsCalculationJob from OVERDUE and PARTIAL schedules (sum of amount_due – amount_paid + penalty; max overdue_days).

NPL numerator / denominator

Element	Definition in ReportsService.getNplReport
NPL loan	DEFAULTED OR (ACTIVE|DELINQUENT AND days_in_arrears ≥ threshold; default 90)
Numerator	SUM(balance) of NPL loans only
Denominator	SUM(balance) of ACTIVE + DELINQUENT + DEFAULTED
Excluded from book	COMPLETED, CANCELLED, PENDING_APPROVAL (not in status filter)
Tenant filter	l.tenant_id = $1 from JWT tenantId
Branch filter	Optional branchId query param

Reversed payments: daily collection and accountability exclude REVERSED. Settled loans COMPLETED are outside NPL active-book denominator. Reschedule-specific rules: not separately modelled beyond schedule status — flag for management if required.

Boundary tests executed (this environment)

Script: backend/scripts/test-npl-classification.js — **VERIFIED** 8/8 PASS including 89/90/91 day ACTIVE boundaries, DEFAULTED always NPL, COMPLETED not NPL by rule, ratio math, aging buckets.

4. Phase 4 — Security review (endpoint inspection)

Control	Finding	Class
JWT + Roles on reports	@UseGuards(JwtAuthGuard, RolesGuard); @Roles admin/manager/superadmin	VERIFIED in source
Tenant on reports	req.user.tenantId passed into service SQL	VERIFIED in source
Helmet + ValidationPipe	main.ts uses helmet() and global ValidationPipe	VERIFIED in source
Throttler	Module present in dependencies/app wiring	IMPLEMENTED BUT NOT VERIFIED
Nav hide = security	Layout role filter is UI only	VERIFIED risk documented
Penetration test	Not performed	NOT IMPLEMENTED

5. Phase 5 — Automated testing

Added pure classification util and node assert script (no new framework). Full Jest suite for payment allocation and Excel buffers: **NOT IMPLEMENTED**. Frontend E2E: **NOT IMPLEMENTED**.

6. Phase 6 — Migrations and deploy

Migration 0029 is additive (IF NOT EXISTS columns on loan_alerts) with down() dropping those columns. Seed scripts are designed idempotent (WHERE NOT EXISTS patterns). Production seed without approval: do not run. Railway config present; production build not executed here — **BLOCKED** on this agent host for full Railway deploy proof.

7. Outstanding risks

• NPL depends on ArrearsCalculationJob running so days_in_arrears is current.
• Management must confirm 90-day threshold if policy differs.
• Incomplete DNA pass on operational forms increases UI inconsistency risk.
• Thin automated coverage outside pure NPL helpers.

8. Deploy / rollback

Deploy: merge source → backend npm i → migration:run → start:dev → frontend npm i → npm run dev. Optional local seeds only. Rollback: restore prior files; migration 0029 down only if needed.

End — HandzJ verification report. No unverified production-success claims.